

Daniel Xu

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EDUCATION

Georgia Institute of Technology

Atlanta, Georgia

B.S. in Mechanical Engineering

Expected Graduation, Spring 2028

- GPA: 3.96; Faculty Honors | National Academy of Engineering Grand Challenges Scholar | Georgia Tech Ice Hockey Team

EXPERIENCE

SANY America

Peachtree City, Georgia

Engineering Intern

May 2026 – Jul 2026

- Built SANY's first Total Cost of Ownership model from scratch in Excel, covering depreciation, financing, fuel, labor, and maintenance across all 30+ machines in the construction equipment line.
- Troubleshoot engine sensors and validated machine telematics reaching company servers; analyzed warranty claim trends to ground maintenance and repair assumptions in field data.
- Reproduced EquipmentWatch's black-box residual-value model with a six-curve nonlinear least-squares fit ($R^2 > 0.9999$), porting Excel Solver to a server-side fitter.
- Grew the assigned calculator into a full sales platform (Next.js, TypeScript, PostgreSQL): live cost breakdowns, competitive comparisons, branded PDF reports, and an interactive lift chart derating bucket capacity by zone.
- Consolidated conflicting machine data held separately by each department into a single governed source of truth with role-based permissions and data-health flagging; now piloting with regional business managers.

Intelligent Vision and Automation Lab, Georgia Tech

Atlanta, Georgia

Undergraduate Researcher

Jan 2026 – Present

- Rearchitected the Arena 3.0 pedestrian simulator from a pure social force model to A* global planning over the environment occupancy map with force-based local steering.
- Restored benchmark validity: the original model let agents pass through walls, ignore the robot, and reverse heading up to four times per second in corridors; verified the fix against trajectory inspection and benchmarking results.
- Generated 20+ static and dynamic environments for evaluating 8+ navigation planners in Arena-Rosnav; co-author on the resulting benchmarking paper (forthcoming).

Laboratory for Intelligent Decision and Autonomous Robots, Georgia Tech

Atlanta, Georgia

Undergraduate Researcher

Aug 2025 – Present

- Adapted a lab-developed 2D dynamic-gap planner to 3D sensing on a Unitree Go2, collapsing the LiDAR ego-cylinder into a 2D ego-circle through a fixed vertical band at the robot's collision envelope.
- Ran the scan-to-command pipeline end to end in Gazebo and in live testing on the physical robot (ROS 2 Humble).
- Designed and 3D-printed all third-party sensor and compute mounting hardware for the Go2 in SolidWorks, covering onboard computer, battery, and cameras, plus a LiDAR mount replacing the stock lid for unobstructed field of view.

PROJECTS

ME 2110 Design Competition, Georgia Tech

Atlanta, Georgia

Design Lead, Four-Robot Autonomous Round

Feb 2026 – Apr 2026

- Designed and built a robot addressing every scoring task from two DC motors and unpowered mechanisms, multiplexing each motor across multiple subsystems and offloading the rest to gravity, elastic tension, and the field's own rotation.
- Built a tape-measure mast to raise the payload up the central spire, removing its coil spring to satisfy a stored-energy rule that named tape measures as banned; upheld after instructor review.
- Designed an unpowered sorter, unfolded by the slide release and fed by the rotating field, that separated pieces by height into elastically tensioned arms released at T-5 s to deposit and clear opposing targets.

Canadian Satellite Design Challenge

Drumheller, Alberta

Team Lead

Jan 2023 – Apr 2023

- Built a can-satellite that collects atmospheric data and transmits it periodically to a Yagi antenna ground station.
- Designed a controlled landing sequence with a parachute cutaway mechanism and flip-out drone propellers, triggered by emulating receiver protocol from the onboard microcontroller to alter flight controller settings.

SKILLS

Programming: Python, C++, C#, Java, MATLAB, TypeScript/JavaScript, R, Arduino/ROBOTC

Software & Tools: SolidWorks, Fusion 360, CATIA, Onshape, ROS2, Gazebo, RViz, Linux, Git, React/Next.js, PostgreSQL, Excel

Engineering: GD&T, DFM, BOM, Rapid Prototyping, CNC & Laser Cutting, Cost Modeling, Technical Writing

Languages: English, Spanish, Chinese (Mandarin)